

Data Assimilation in Glaciology

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Abstract: Data assimilation has always been a particularly active area of research in glaciology. While many properties at the surface of glaciers and ice sheets can be directly measured from remote sensing or in situ observations (surface velocity, surface elevation, thinning rates, etc.), many important characteristics, such as englacial and basal properties, as well as past climate conditions, remain difficult or impossible to observe. Data assimilation has been used for decades in glaciology in order to infer unknown properties and boundary conditions that have important impact on numerical models and their projections. The basic idea is to use observed properties, in conjunction with ice flow models, to infer these poorly known ice properties or boundary conditions. There is, however, a great deal of variability among approaches. Constraining data can be of a snapshot in time, or can represent evolution over time. The complexity of the flow model can vary, from simple descriptions of lubrication flow or mass continuity to complex, continent-wide Stokes flow models encompassing multiple flow regimes. Methods can be deterministic, where only a best fit is sought, or probabilistic in nature. We present in this chapter some of the most common applications of data assimilation in glaciology, and some of the new directions that are currently being developed.

6.1 Background and definitions

6.1.1 Observations of Ice Sheets and Glaciers

A *glacier* can be defined as a persistent body of ice that flows and deforms under its own weight. There are approximately 160,000 glaciers on Earth ranging in size from ~ 1 to thousands of square kilometres. An *ice sheet* is a body of glacial ice over 50,000 km² in area. There are only two surviving ice sheets today: Antarctica and Greenland.

For over half a century, scientists have recorded comprehensive quantitative measurements of the shape and speed of glaciers and ice sheets – to the point where currently satellite measurements are made of the elevation and surface velocity of glaciers and ice sheets with high accuracy and resolution. These data complement lower-quality satellite observations made in previous decades. Geophysical and airborne observing platforms

are additionally able to capture the shape of internal reflecting layers, which encapsulate ice-sheet history, as well as the ice thickness.

Many of the methods discussed in this chapter rely on data which are spatially (and in some cases temporally) extensive, and current glaciological remote sensing methods provide a wealth of such information.

6.1.2 Modelling of Ice Sheets and Glaciers

Assimilation of the vast glaciological data collected requires physically based models. The flow of glaciers and ice sheets is modelled by solving conservation laws: conservation of momentum (stress balance), conservation of mass (continuity equation), and conservation of energy. For simplicity, we only describe here the first two conservation laws because the thermal regime of the ice is rarely directly involved in data assimilation methods.

Stress Balance

The stress balance of ice is generally modelled as an incompressible Stokes flow:

$$\begin{cases} \nabla \cdot \boldsymbol{\sigma}' - \nabla P + \rho \mathbf{g} = \mathbf{0} \\ \nabla \cdot \mathbf{v} = 0, \end{cases} \quad (6.1)$$

where $\boldsymbol{\sigma}'$ is the deviatoric stress tensor, P is the ice pressure, and $\mathbf{g}$ is the acceleration due to gravity. While ice may behave elastically over short time scales (e.g. tidal cycle), its long-term behaviour is described by a viscous constitutive law:

$$\boldsymbol{\sigma}' = 2\mu \dot{\boldsymbol{\epsilon}}, \quad (6.2)$$

where the ice viscosity, μ , follows a Norton–Hoff viscoplastic law referred to as Glen's flow law (Glen, 1955):

$$\mu = \frac{B}{2 \dot{\epsilon}_e^{(n-1)/n}}, \quad (6.3)$$

and where B is the ice viscosity factor ($A = B^{-n}$, the ice rate factor, is sometimes used instead of B), n is Glen's exponent, generally taken as 3, and $\dot{\epsilon}_e$ is the effective strain rate.

In terms of boundary conditions, it is generally assumed that the ice–air interface is stress free: $\boldsymbol{\sigma} \cdot \mathbf{n} = \mathbf{0}$, and hydrostatic water pressure is applied at the ice–ocean interface:

$$\boldsymbol{\sigma} \cdot \mathbf{n} = \rho_w g z \mathbf{n}, \quad (6.4)$$

where $\mathbf{n}$ is the outward pointing unit vector, ρ_w is the density of seawater, and z is the vertical coordinate (zero at sea level). Finally, at the ice–bed interface, we have a non-penetration condition ($\mathbf{v} \cdot \mathbf{n} = 0$, if there is no basal melt) and basal friction is applied along the tangential plane. To remain general, we write this friction condition as:

$$\boldsymbol{\tau}_b = \boldsymbol{\sigma} \cdot \mathbf{n} - (\mathbf{n} \cdot \boldsymbol{\sigma} \cdot \mathbf{n}) \mathbf{n} = -C f_\tau(|\mathbf{v}_b|, N) \mathbf{v}_b, \quad (6.5)$$

where $C \geq 0$ is a friction coefficient and f_τ is a function of the basal velocity, $\mathbf{v}_b$, and basal effective pressure, N .

Since solving Stokes flow at large scale can be computationally expensive, approximations are sometimes used. The simplest is the *Shallow-Ice Approximation*, which assumes that ice motion is determined solely by vertical shearing stress, and is appropriate for slow-moving glaciers and ice-sheet interiors (Greve and Blatter, 2009). The *Shallow-Shelf Approximation* (SSA; sometimes *Shelfy-Stream Approximation*) considers flow to be vertically uniform and controlled by membrane stresses (MacAyeal, 1989), and applies to fast-flowing ice streams and ice shelves. The *First-Order Approximation* includes vertical shearing as well as membrane stresses but neglects ‘bridging’ (non-hydrostatic) stresses (Blatter, 1995; Pattyn, 2003). The majority of continental-scale ice-sheet modelling studies implement one or a combination of these approximations.

Mass Transport

Depth integrating the continuity equation (the second equation of Eq. (6.1)) and applying the necessary boundary conditions at the base and surface of the ice sheet leads to the two-dimensional equation of mass transport, which governs the change in geometry of the ice sheet:

$$\frac{\partial H}{\partial t} = -\nabla \cdot H \bar{\mathbf{v}} + \dot{m}_s - \dot{m}_b, \quad (6.6)$$

where H is the ice thickness, $\bar{\mathbf{v}}$ is the depth-averaged velocity, $\dot{m}_s$ is the surface mass balance (positive when snow accumulation), and $\dot{m}_b$ is the melt rate at the base of the ice (positive when melting).

6.1.3 Model-Data Assimilation

We begin by defining what is meant by ice-sheet data assimilation. Ice-sheet data assimilation combines models of ice-sheet and glacier physics with observations in order to infer unresolved or unobservable physical properties. The aim of ice-sheet data assimilation varies by implementation: in some cases, the properties

themselves are of direct interest, whereas in other cases the assimilation is a calibration procedure for model projections. Ice-sheet data assimilation is an extremely diverse topic, with the types of data assimilated ranging from internal radar layers to satellite altimetry, and with models ranging from simple mass-balance models to sophisticated Stokes flow. In order to establish a common language for discussing these different types of assimilation, we make some formal definitions. Broadly, ice-sheet data assimilation can be described by the following optimisation problem: find $\boldsymbol{\alpha}$ such that:

$$\boldsymbol{\alpha} = \arg \min_{\mathbf{x}=f(\boldsymbol{\alpha})} \mathcal{J}(\boldsymbol{\alpha}), \quad (6.7)$$

$$\mathcal{J}(\boldsymbol{\alpha}) = \frac{1}{2} \int_{\Omega} \frac{(y^{obs} - \mathcal{H}(\mathbf{x}))^2}{\sigma_y^2} d\Omega, \quad (6.8)$$

where Ω is the computational domain of the function f . The functional $\mathcal{J}$ is referred to as a *cost function* (or sometimes an *objective function*), and quantifies the misfit between the model and available observations. The model *state* is $\mathbf{x}$, the solution of the *forward model*, f , for a given $\boldsymbol{\alpha}$. The minimisation is carried out over all possible values of $\boldsymbol{\alpha}$ – which is a parameter, or set of parameters, for the model. For instance, $\boldsymbol{\alpha}$ might be the surface mass balance over a finite historical period, or it might be the set of all values describing the basal friction field under an ice sheet. These parameters are often referred to as the *controls* of the assimilation. Here, y^{obs} describes a spatial field of an ice-sheet *observable* (e.g. surface elevation), and Ω is the model domain. Here σ_y represents the uncertainty in y^{obs} , and $\mathcal{H}$ is the *observation operator* that maps the model state, $\mathbf{x}$, to the space of observables.

Eqs. (6.7) and (6.8) are meant as representative of the general form of the assimilation, but can vary depending on the approach. Here y^{obs} is written as a scalar-valued function, but the observable can be a vector-valued function (e.g. ice velocity). The integral in Eq. (6.8) is a spatial integral but integration can be over time as well. The parameters $\boldsymbol{\alpha}$ are in general spatially (and temporally) varying but by convention are represented by a discrete vector. While many assimilation problems use the integral formulation in Eq. (6.8), $\mathcal{J}$ is sometimes constructed as a discrete sum over spatiotemporal observation points i :

$$\mathcal{J}(\boldsymbol{\alpha}) = \frac{1}{2} \sum_i \frac{(y_i^{obs} - \mathcal{H}_i(\mathbf{x}))^2}{\sigma_i^2}. \quad (6.9)$$

As shown in Section 6.3.3, the distinction between Eqs. (6.8) and (6.9) is critical to Bayesian interpretation of uncertainties.

Strictly, Eq. (6.7) describes *variational* data assimilation (VDA), in which optimal fit to observations is found subject

to model physics. VDA is the approach most widely used in glaciological modelling, although other types are discussed in this chapter as well. The form of Eq. (6.7) is intentionally simplistic and, in practice, cost functions can differ in various ways, as discussed in this and later sections.

6.1.4 Regularisation

In many cases, the observables in an assimilation framework are insensitive to small variations in the controls. The inverse problem as defined by Eq. (6.7) is typically *ill posed*: there may be many solutions α to the minimisation problem (Hadamard, 1902), or the problem may be ill conditioned (i.e. sensitive to small uncertainties in the observational data). A common strategy to address this problem is to add a *regularisation* term to the cost function, which penalises non-physical behaviours. For instance, a common approach in estimating ice-sheet basal friction is to use Tikhonov regularisation (Morlighem et al., 2010), which penalises the square gradient of the control field:

$$\mathcal{J}(\alpha) = \underbrace{\frac{1}{2} \int_{\Omega} \frac{(y^{obs} - \mathcal{H}(\mathbf{x}))^2}{\sigma_y^2} d\Omega}_{\text{misfit, } \mathcal{J}_0} + \underbrace{\frac{1}{2} \int_{\Omega} \gamma |\nabla \alpha|^2 d\Omega}_{\text{regularisation}}, \quad (6.10)$$

where γ is a parameter that sets the level of regularisation. The first term, $\mathcal{J}_0$, is often referred to as the *misfit* or *data* norm, while the integral in the second term is sometimes referred to as the *model* norm (Waddington et al., 2007). The minimisation of the cost function $\mathcal{J}$ is thus a trade-off between minimising the two. Careful selection of the regularisation parameter γ is important. An overly strong regularisation can lead to poor model-data fit, while overly weak regularisation can lead to non-physical features in the optimal controls, or over-fitting. A commonly used approach to identify optimal parameters is the *L-curve* analysis (Hansen, 2000; Gillet-Chaulet et al., 2012), which examines the relationship between misfit and model norm as the regularisation parameter is varied. In approaches with more than one control field (or more than one regularisation parameter) an *L-surface* may be used (Furst et al., 2015). There are alternative methods of finding the appropriate regularisation, however, such as the discrepancy principle (Habermann et al., 2013) or the Lagrange multiplier method of Waddington et al. (2007).

The regularisation term is sometimes referred to as a ‘prior’ (Barnes et al., 2021), which lends itself to a Bayesian interpretation of assimilation (see Section 6.3.3). In this sense, the regularisation is imposing *a priori* knowledge regarding the controls of the assimilation. Care should be taken, however, not to impose knowledge that one does not possess as this could influence quantifications of uncertainty (Arthern, 2015).

6.1.5 Cost Function Minimisation

Equation (6.7) represents, in most glaciological assimilations, a very large-scale non-linear optimisation (or rather, minimisation) problem. Methods to solve this minimisation problem have included ad-hoc approaches (Chandler et al., 2006; Price et al., 2011) or explicit reduction of the control dimension to create a more manageable problem (Payne et al., 2004) among other methods. However, by far the most common approach taken is an iterative one in which search directions are updated based on cost function gradients (MacAyeal, 1993). Such approaches include simple gradient-descent methods (e.g. Joughin et al., 2004) and quasi-Newton methods (e.g. Gillet-Chaulet et al., 2012; Barnes et al., 2021). Quasi-Newton methods, such as the Limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm (L-BFGS, Gilbert and Lemaréchal, 1989; Nocedal and Wright, 2006), construct a low-rank approximation to the *Hessian* (or second derivative matrix) of the cost function through consecutive gradient calculations, and generally converge faster than simple gradient descent. ‘True’ Newton methods, which use the exact Hessian of the cost function, are expected to converge faster than quasi-Newton methods, but to date such methods have not been used for large-scale glaciological data assimilation, predominantly because of the computational cost in finding the inverse of the Hessian (see Section 6.3). Recently however, a study has made use of a more sophisticated Hessian approximation (Shapero et al. (2021); see also Section 6.3.6), which converges faster than L-BFGS in experiments carried out.

Methods for calculating forward model gradients vary greatly across various assimilation frameworks. A common approach is the *adjoint* method, which solves the mathematical adjoint of the linearised model (see Section 6.2.2). Such methods are advantageous as they require a single adjoint solution, as opposed to individually perturbing potentially tens of thousands of parameters (MacAyeal, 1992a). While most approaches to date derive the continuous adjoint model, which is then discretised, there are an increasing number of studies making use of an *algorithmic differentiation* (AD; Heimbach and Bugnion, 2009). AD software tools generate a discretised form of the adjoint model directly from the discretised forward model through repeated application of the chain rule at the level of the numerical code (Griewank and Walther, 2008). In the following sections, different applications of these approaches are examined.

6.2 Inferring Ice Dynamical Properties and Boundary Conditions

In this section, we describe some of the popular data assimilation techniques that have been employed to constrain poorly known boundary conditions and ice material

properties from surface observations. We first describe ad hoc methods that are easy to implement but require running the model over long time scales and assume that the system is in steady state. We then present the time-independent (or snapshot) variational approaches that have been particularly popular in glaciology, and then the time-dependent approaches that are also based on variational methods.

6.2.1 Ad Hoc Methods

Inference from Surface Speed

Described in Price et al. (2011), this method is used to infer basal drag from surface speed. The approach starts with a very large value for the basal drag (i.e. no sliding), as described in Eq. (6.5). The model is then run forward while keeping the geometry fixed, until the ice temperature, velocity, and viscosity reach a steady state. Once a steady state is reached, the basal stress $|\tau_b|$ is evaluated. The modelled surface speed is then subtracted from the observed surface speed to estimate the sliding speed, v_b , that was previously assumed to be 0. These sliding speeds, together with the inferred basal stress, are used to determine the sliding coefficient by solving:

$$C = \frac{|\tau_b|}{f_\tau(|v_b|, N)|v_b|}. \quad (6.11)$$

Using this new distribution of C , the model is run again to steady state. The evaluation of C by determining the basal drag, $|\tau_b|$, and adding the misfit between modelled and observed surface velocities to the sliding velocities, is repeated until convergence is reached.

To our knowledge, this method has not been used outside this study, but is easy to implement and relies on the simple concept that decreasing the friction coefficient has a direct, positive impact on the surface velocities.

Inference from Surface Height

Pollard and DeConto (2012) proposed a similar method but using surface height, s , as a target instead of surface velocities. The idea is to run a model forward in time and adjust the friction coefficient, C , so that the surface heights match observed elevations, assuming that the ice sheet is in steady state. In regions where the modelled surface height is too low, friction is increased, which allows for the ice to thicken, whereas in regions where the ice is too high, friction is decreased to allow for more sliding and thinning.

In practice, the friction is updated every 5,000 years as follows:

$$C_{\text{new}} = C_{\text{old}} \times 10^{\Delta z}, \quad (6.12)$$

where:

$$\Delta z = \max(-1.5, \min(1.5, (s_{\text{obs}} - s)/s_0)), \quad (6.13)$$

where $s_0 = 500$ m is a scaling parameter. This method allows for the modelled height to reach an average of 50 m of misfit between the model and observation within 200,000 to 400,000 years. While simple, it still requires running the model for a long period of time and is difficult to apply to more sophisticated ice sheet models that typically only run for a few centuries.

6.2.2 Time-Independent/Snapshot Variational Approaches

One of the limitations of these ad hoc methods is that they require the model to be in steady state, which is not ideal when studying rapidly changing systems. A popular alternative approach that removes this assumption was introduced by MacAyeal (1992b, 1993) in the 1990s, and has been applied almost exclusively, until relatively recently, to the two-dimensional SSA. Known as a ‘control method’, it is a direct application of the optimal control theory to glaciology (Bryson and Ho, 2018). A type of VDA, the method consists of minimising the cost function through a gradient-descent approach. We first describe VDA in a general context, for both discrete and continuous model equations, and then show how it is applied to the SSA.

Discrete Adjoint Method

In the discrete form, the model state is a vector $\mathbf{x} \in \mathbb{R}^m$, and the control is $\mathbf{a} \in \mathbb{R}^n$. We write $J(\mathbf{x}, \mathbf{a}) \in \mathbb{R}_+$ the cost function that we want to minimise, and define $\mathcal{J}(\mathbf{a}) = J(\mathbf{x}(\mathbf{a}), \mathbf{a})$ such that $\mathbf{x}(\mathbf{a})$ satisfies the model equation $\mathbf{g}(\mathbf{x}, \mathbf{a}) = \mathbf{0}$, which is an m -dimensional zero vector. For instance, if one uses the finite-element method, we would have $\mathbf{g}(\mathbf{x}, \mathbf{a}) = \mathbf{K}\mathbf{x} - \mathbf{F}$, where $\mathbf{K}$ is the stiffness matrix and $\mathbf{F}$ the load vector (where $\mathbf{K}$ and $\mathbf{F}$ in general depend on the control $\mathbf{a}$). We would like to find the gradient of the cost function in order to minimise it, as described in Section 6.1.5.

The most intuitive approach to compute this gradient would be to apply a small perturbation to each of the components of the control, $\mathbf{a}$ using a finite-difference method to construct the gradient of the cost function. This method, however, requires to solve n times the model equations, $\mathbf{g}(\mathbf{x}, \mathbf{a}) = \mathbf{0}$. While simple, this approach becomes very inefficient and computationally prohibitive when $n \gtrsim 10$. Variational data assimilation allows to compute this gradient by solving only two sets of equations. The gradient of the cost function is:

$$\nabla \mathcal{J} = \frac{\partial J}{\partial \mathbf{a}} + \frac{\partial J}{\partial \mathbf{x}} \frac{\partial \mathbf{x}}{\partial \mathbf{a}}. \quad (6.14)$$

Since J has an explicit expression, the first two vectors are easy to compute. The Jacobian matrix, $\partial \mathbf{x} / \partial \mathbf{a} \in \mathbb{R}^{n \times m}$, however, is difficult to evaluate, especially if n and m are large. If we take the derivative of the forward model with respect to the control, $\mathbf{a}$, we find:

$$\frac{d\mathbf{g}}{d\alpha} = \frac{\partial \mathbf{g}}{\partial \alpha} + \frac{\partial \mathbf{g}}{\partial \mathbf{x}} \frac{\partial \mathbf{x}}{\partial \alpha} = \mathbf{0}, \quad (6.15)$$

which is $\mathbf{0}$ since the forward model must be satisfied. Using this equation, we can rewrite the problematic term as:

$$\frac{\partial \mathbf{x}}{\partial \alpha} = - \left[\frac{\partial \mathbf{g}}{\partial \mathbf{x}} \right]^{-1} \frac{d\mathbf{g}}{d\alpha}. \quad (6.16)$$

Substituting this expression in Eq. (6.14) gives:

$$\nabla \mathcal{J} = \frac{\partial J}{\partial \alpha} - \frac{\partial J}{\partial \mathbf{x}} \left[\frac{\partial \mathbf{g}}{\partial \mathbf{x}} \right]^{-1} \frac{d\mathbf{g}}{d\alpha}. \quad (6.17)$$

We define the *adjoint state*, $\lambda \in \mathbb{R}^m$, as:

$$\lambda^T = - \frac{\partial J}{\partial \mathbf{x}} \left[\frac{\partial \mathbf{g}}{\partial \mathbf{x}} \right]^{-1}, \quad (6.18)$$

or, written differently:

$$\left[\frac{\partial \mathbf{g}}{\partial \mathbf{x}} \right]^T \lambda = - \left[\frac{\partial J}{\partial \mathbf{x}} \right]^T, \quad (6.19)$$

which is generally referred to as the *adjoint model*. The calculation of the adjoint state requires solving a system of equations which has the same size as the forward model. Once the adjoint state is calculated, we can compute the gradient of the cost function as:

$$\nabla \mathcal{J} = \frac{\partial J}{\partial \alpha} + \lambda^T \frac{d\mathbf{g}}{d\alpha}. \quad (6.20)$$

It can be shown that the adjoint state is equivalent to the vector of Lagrange multipliers introduced to enforce that the initial model equation, or forward model, is satisfied.

Continuous Adjoint Method

If we consider a set of continuous model equations, we can follow the same steps as in the previous section, but this time we introduce a Lagrangian to enforce the constraint that the model equations need to be satisfied. The model state, $\mathbf{x}$, and the control, α , are now fields in the appropriate Banach spaces (note that, for simplicity and brevity, we will assume that all variables are defined in the appropriate space). The objective function is $J(\mathbf{x}, \alpha) \in \mathbb{R}_+$ and, again, we define $\mathcal{J}(\alpha) = J(\mathbf{x}(\alpha), \alpha)$ such that $\mathbf{x}(\alpha)$ satisfies the model equation $\mathbf{g}(\mathbf{x}, \alpha) = \mathbf{0}$.

We want to find α such that it minimises the cost function. The idea of optimal control is to introduce a Lagrangian for this optimisation problem:

$$\mathcal{L}(\mathbf{x}, \lambda, \alpha) = J(\mathbf{x}, \alpha) + \int_{\Omega} \lambda \cdot \mathbf{g}(\mathbf{x}, \alpha) d\Omega. \quad (6.21)$$

Similarly, the new variable, λ , is the Lagrange multiplier (or adjoint state) for the constraint $\mathbf{g}(\mathbf{x}, \alpha) = \mathbf{0}$. If $\mathbf{x}$ is solution of the forward model, then

$$\mathcal{J}(\alpha) = \mathcal{L}(\mathbf{x}, \lambda, \alpha) \quad (6.22)$$

since, in this case, $\mathbf{g}(\mathbf{x}, \alpha) = \mathbf{0}$. If we take the directional derivative of this expression with respect to the control along the direction β , and apply the chain rule, this yields:

$$\langle \nabla \mathcal{J}, \beta \rangle = \left\langle \frac{\partial \mathcal{L}}{\partial \mathbf{x}}, \mathbf{x}'(\beta) \right\rangle + \left\langle \frac{\partial \mathcal{L}}{\partial \lambda}, \lambda'(\beta) \right\rangle + \left\langle \frac{\partial \mathcal{L}}{\partial \alpha}, \beta \right\rangle. \quad (6.23)$$

Because we want the forward model to be satisfied, the second term on the right-hand side is zero:

$$\forall \mu, \left\langle \frac{\partial \mathcal{L}}{\partial \lambda}, \mu \right\rangle = \langle \mu, \mathbf{g}(\mathbf{x}, \alpha) \rangle = 0. \quad (6.24)$$

If we choose the adjoint state such that the Lagrangian is also stationary along $\mathbf{x}$, the first term is zero:

$$\forall \mathbf{w}, \left\langle \frac{\partial \mathcal{L}}{\partial \mathbf{x}}, \mathbf{w} \right\rangle = \left\langle \frac{\partial J(\mathbf{x}, \alpha)}{\partial \mathbf{x}}, \mathbf{w} \right\rangle + \left\langle \lambda, \frac{\partial \mathbf{g}}{\partial \mathbf{x}} \{\mathbf{w}\} \right\rangle = 0, \quad (6.25)$$

where the $\{\cdot\}$ indicates the action of the differential operator in a given direction. Using the properties of inner products, this can be rearranged:

$$\left\langle \left(\frac{\partial \mathbf{g}}{\partial \mathbf{x}} \right)^T \{\lambda\}, \mathbf{w} \right\rangle = - \left\langle \frac{\partial J(\mathbf{x}, \alpha)}{\partial \mathbf{x}}, \mathbf{w} \right\rangle, \quad (6.26)$$

yielding the *adjoint equations* to be solved for λ . This then provides a simple expression for the gradient of the cost function with respect to the control:

$$\langle \nabla \mathcal{J}, \beta \rangle = \left\langle \frac{\partial \mathcal{L}}{\partial \alpha}, \beta \right\rangle = \left\langle \frac{\partial J}{\partial \alpha}, \beta \right\rangle + \left\langle \lambda \cdot \frac{\partial \mathbf{g}}{\partial \alpha}(\mathbf{x}, \alpha), \beta \right\rangle \quad (6.27)$$

and we identify the gradient as:

$$\nabla \mathcal{J} = \frac{\partial J}{\partial \alpha} + \lambda \cdot \frac{\partial \mathbf{g}}{\partial \alpha}(\mathbf{x}, \alpha). \quad (6.28)$$

Application to SSA

Here we apply this approach to the Shallow-Shelf Approximation. The equations of SSA are derived from the full-Stokes equations (Eq. (6.1)), and rely on two assumptions. The first one is that there is no bridging effect, which means that the vertical normal stress is purely lithostatic, or equal to the weight of ice vertically above. The second assumption is that vertical shear is negligible, so that velocities are assumed depth-independent. These two assumptions lead to the following depth-integrated model:

$$\left\{ \begin{aligned} \frac{\partial}{\partial x} \left(4H\mu \frac{\partial v_x}{\partial x} + 2H\mu \frac{\partial v_y}{\partial y} \right) + \frac{\partial}{\partial y} \left(H\mu \frac{\partial v_x}{\partial y} + H\mu \frac{\partial v_y}{\partial x} \right) &= \rho g H \frac{\partial s}{\partial x} + C v_x \\ \frac{\partial}{\partial x} \left(H\mu \frac{\partial v_x}{\partial y} + H\mu \frac{\partial v_y}{\partial x} \right) + \frac{\partial}{\partial y} \left(4H\mu \frac{\partial v_y}{\partial y} + 2H\mu \frac{\partial v_x}{\partial x} \right) &= \rho g H \frac{\partial s}{\partial y} + C v_y \end{aligned} \right\}, \quad (6.29)$$

where we assume here, for simplicity, that the friction law is $\tau_b = -C \mathbf{v}$.

MacAyeal (1992b) applied an adjoint-based inversion to infer the basal friction under Ice Stream E, Antarctica. Joughin et al. (2004) modified the method to determine the friction under Ross Ice Stream assuming a weak plastic bed. Rommelaere and MacAyeal (1997) extended this approach to infer the ice rheological parameter, B , of the Ross ice shelf, Antarctica, with the same Shallow-Shelf model, from surface velocities. Larour (2005) applied a similar method to infer B of the Ronne ice shelf. This method was then extended to higher-order and full-Stokes models by Morlighem et al. (2010) and applied to the entire Antarctic ice sheet (Morlighem et al., 2013b). It is now used by many glaciologists and available in several ice sheet models (Elmer, ISSM, STREAMICE, U'a, BISICLES, MALI, etc).

As an illustration, we describe here how to derive the gradient of a classical cost function:

$$J(\mathbf{v}) = \frac{1}{2} \int_{\Omega} (v_x - v_x^{obs})^2 + (v_y - v_y^{obs})^2 d\Omega. \quad (6.30)$$

In this case, the observation operator is the identity function since we observe the model state itself, $\mathbf{x} = (v_x, v_y)^T$. Let's assume that we are using the SSA equations with homogeneous Dirichlet conditions along the boundary and we are inferring the basal friction coefficient, $\alpha = C$, from observed surface velocities. Let's further ignore the dependency of the ice viscosity, μ , on the ice velocity in the computation of the adjoint state (which is a common assumption in glaciology; see, for example, MacAyeal, 1992a). Using the continuous approach (Section 6.2.2), the adjoint equations (Eq. (6.26)) are: $\forall \mathbf{w}$

$$\begin{aligned} \left\langle \frac{\partial J(\mathbf{v}, \alpha)}{\partial \mathbf{v}}, \mathbf{w} \right\rangle + \left\langle \lambda, \frac{\partial \mathbf{g}}{\partial \mathbf{v}} \{\mathbf{w}\} \right\rangle &= \int_{\Omega} (v_x - v_x^{obs}) w_x \\ &+ (v_y - v_y^{obs}) w_y d\Omega \\ &+ \int_{\Omega} \lambda_x \frac{\partial}{\partial x} \left(4H\mu \frac{\partial w_x}{\partial x} + 2H\mu \frac{\partial w_y}{\partial y} \right) \\ &+ \lambda_x \frac{\partial}{\partial y} \left(H\mu \frac{\partial w_x}{\partial y} + H\mu \frac{\partial w_y}{\partial x} \right) - C w_x \lambda_x d\Omega \\ &+ \int_{\Omega} \lambda_y \frac{\partial}{\partial x} \left(H\mu \frac{\partial w_x}{\partial y} + H\mu \frac{\partial w_y}{\partial x} \right) \\ &+ \lambda_y \frac{\partial}{\partial y} \left(4H\mu \frac{\partial w_y}{\partial y} + 2H\mu \frac{\partial w_x}{\partial x} \right) - C w_y \lambda_y d\Omega = 0. \end{aligned} \quad (6.31)$$

If we integrate the two last lines by parts twice, and assume homogeneous boundary conditions, this equation becomes:

$$\begin{aligned} &\int_{\Omega} (v_x - v_x^{obs}) w_x + (v_y - v_y^{obs}) w_y d\Omega \\ &+ \int_{\Omega} w_x \frac{\partial}{\partial x} \left(4H\mu \frac{\partial \lambda_x}{\partial x} + 2H\mu \frac{\partial \lambda_y}{\partial y} \right) \\ &+ w_x \frac{\partial}{\partial y} \left(H\mu \frac{\partial \lambda_x}{\partial y} + H\mu \frac{\partial \lambda_y}{\partial x} \right) - C w_x \lambda_x d\Omega \\ &+ \int_{\Omega} w_y \frac{\partial}{\partial y} \left(4H\mu \frac{\partial \lambda_y}{\partial y} + 2H\mu \frac{\partial \lambda_x}{\partial x} \right) \\ &+ w_y \frac{\partial}{\partial x} \left(H\mu \frac{\partial \lambda_x}{\partial y} + H\mu \frac{\partial \lambda_y}{\partial x} \right) - C w_y \lambda_y d\Omega = 0, \end{aligned} \quad (6.32)$$

and since this is true for all $\mathbf{w}$, the adjoint equations are:

$$\begin{cases} \frac{\partial}{\partial x} \left(4H\mu \frac{\partial \lambda_x}{\partial x} + 2H\mu \frac{\partial \lambda_y}{\partial y} \right) + \frac{\partial}{\partial y} \left(H\mu \frac{\partial \lambda_x}{\partial y} + H\mu \frac{\partial \lambda_y}{\partial x} \right) - C \lambda_x = -(v_x - v_x^{obs}) \\ \frac{\partial}{\partial y} \left(4H\mu \frac{\partial \lambda_y}{\partial y} + 2H\mu \frac{\partial \lambda_x}{\partial x} \right) + \frac{\partial}{\partial x} \left(H\mu \frac{\partial \lambda_x}{\partial y} + H\mu \frac{\partial \lambda_y}{\partial x} \right) - C \lambda_y = -(v_y - v_y^{obs}). \end{cases} \quad (6.33)$$

The adjoint state can therefore be solved using very similar equations as the forward model but with a different right-hand side. Once the adjoint state is calculated, the gradient of the cost function is simply:

$$\nabla J = -\lambda \cdot \mathbf{v}. \quad (6.34)$$

A similar approach can be applied to the discretised equations, leading to a similar gradient expression. We must however pay attention to the fact that the discretised gradient provides the sensitivity of the cost function with respect to the value of the control at each node or cell, which is not the same as the continuous gradient. In other words, the discretised gradient for node i is related to the continuous gradient as follows:

$$\nabla J_i = \langle \nabla J, \varphi_i \rangle = \int_{\Omega} -\lambda \cdot \mathbf{v} \varphi_i d\Omega, \quad (6.35)$$

where φ_i is the shape function associated with C_i , the friction coefficient at node i . The discrete and continuous gradients can therefore be converted from one form to the other by using the mass matrix (e.g. Li et al., 2017).

Simultaneous Inference of Multiple Parameter Fields

For reasons of simplicity, the preceding example considers only a single control field, C . However, in many cases, other key spatially varying properties must be inferred, such as the ice viscosity factor, B , which depends on ice temperature (Cuffey and Paterson, 2010) but also on the degree of damage (Borstad et al., 2013). The approach for the Shallow-Shelf model can be easily generalised to account for two parameter fields, as has been done in numerous studies (e.g. Lee et al., 2015; Barnes et al., 2021). Concerns arise in such cases regarding potential compensating effects of the parameters,

which can lead to multiple solutions (Ranganathan et al., 2020), but to date there is not a consensus on whether such ‘dual inversions’ lead to non-uniqueness.

Alternative Approaches

Arthern and Gudmundsson (2010) proposed another method, which is based on reformulating the optimisation as an inverse Robin problem. The forward model (called here the Neumann problem as a Neumann boundary condition is applied at the surface) is modified to solve a Dirichlet problem, where the surface velocity is now imposed as being equal to observed surface speeds. If the model is perfectly calibrated, these two problems should have the same final state. By introducing a specific cost function proposed initially by Kohn and Vogelius (1984):

$$\mathcal{J}(\alpha) = \int_{\Gamma_s} (\mathbf{v}^N - \mathbf{v}^D) \cdot (\boldsymbol{\sigma}^N - \boldsymbol{\sigma}^D) \cdot \mathbf{n} \, d\Gamma, \quad (6.36)$$

where Γ_s is the ice surface where surface speeds are imposed in the Dirichlet problem, and the superscript N and D denote the velocity and stress tensors from the Neumann and Dirichlet problem solutions. Using this approach, if basal friction is assumed to be $\boldsymbol{\tau}_b = -C \mathbf{v}_b$, the gradient of this cost function with respect C is simply

$$\nabla \mathcal{J} = |\mathbf{v}^D|^2 - |\mathbf{v}^N|^2. \quad (6.37)$$

This method can also be employed to find the rheology factor, B . This approach is simple in the sense that it is easy to implement, but is not flexible in terms of choice of cost function.

6.2.3 Time-Dependent Approaches

Using a time-dependent model (i.e. the stress balance coupled with mass transport) provides stronger constraints for assimilating time-dependent data, and expands the types of data that can be used for constraints in the cost function, such as time-evolving altimetry. It also removes the assumption that all observational data are contemporaneous, which must be made for time-independent assimilation and could lead to inaccuracies for highly dynamic systems.

As in time-independent approaches, minimisation of the cost function involves calculating its derivatives with respect to the assimilation controls – but with a time-dependent model the complexity is increased. A continuous adjoint method (Section 6.2.2) can be applied, as is done in Michel et al. (2014), which assimilates transient surface elevation data using a Shallow-Ice model to estimate ice thickness. With non-local stress balances such as Stokes and Shallow-Shelf, however, model complexity makes such approaches challenging. As such, studies which focus on time-

dependent adjoint-based assimilation have started to adopt algorithmic differentiation (AD).

Algorithmic Differentiation

Algorithmic differentiation calculates gradient information for a given cost function and forward model without the need for explicit derivation of the adjoint equations. To illustrate, we define $\mathbf{x}^k$ as the discretised state vector for a given model at time step k . $\mathbf{x}^k$ includes velocity, ice thickness, and possibly additional prognostic variables (such as ice temperature). A model time step can be expressed as

$$\mathbf{x}^{k+1} = F(\mathbf{x}^k, \alpha^k), \quad (6.38)$$

where $F(\cdot, \cdot)$ represents the calculation of velocities and update of thickness (and other prognostic variables) over a single time step. The parameters, α , can, in general, vary per time step (as indicated by the superscript) but can also vary on much longer intervals, or not at all (Goldberg et al., 2015). In the special case where the cost function $\mathcal{J}$ depends only on the final state and the parameters are time-constant (i.e. $\alpha^k \equiv \alpha$), the gradient of the cost function is

$$\begin{aligned} \nabla \mathcal{J} &= \frac{\partial \mathcal{J}}{\partial \mathbf{x}^k} \frac{d\mathbf{x}^k}{d\alpha} \\ &= \frac{\partial \mathcal{J}}{\partial \mathbf{x}^k} \left(\frac{\partial \mathbf{F}}{\partial \alpha} + \frac{\partial \mathbf{F}}{\partial \mathbf{x}^{k-1}} \frac{d\mathbf{x}^{k-1}}{d\alpha} \right) \\ &= \frac{\partial \mathcal{J}}{\partial \mathbf{x}^k} \left(\frac{\partial \mathbf{F}}{\partial \alpha} + \frac{\partial \mathbf{F}}{\partial \mathbf{x}^{k-1}} \left(\frac{\partial \mathbf{F}}{\partial \alpha} + \frac{\partial \mathbf{F}}{\partial \mathbf{x}^{k-2}} \frac{d\mathbf{x}^{k-2}}{d\alpha} \right) \right) \\ &= \dots \end{aligned} \quad (6.39)$$

The expression can be continually expanded using the chain rule for calculus. *Reverse mode* AD implements Eq. (6.39) through repeated left-multiplication of the linear operators. The chain rule is applied not only at the level of a time step but to individual numerical operations within functions and subroutines. It can be shown (Heimbach and Bugnion, 2009) that this computation is mathematically equivalent to solving the adjoint model backward in time. Note that the gradients of $\mathbf{F}$ are evaluated at the state $\mathbf{x}^i$, where $i = k, k-1, \dots$ so solving the adjoint requires either storage or recomputation of the state at each backward time step.

To date, AD has been applied to a number of ice-sheet models for the purpose of time-dependent assimilation. Larour et al. (2014) applied ADOL-C, an AD tool for C++, to a Shallow-Shelf version of the finite-element Ice Sheet System Model (ISSM). The resulting adjoint was used to assimilate surface elevation data derived from ICESat altimetry for the Northeast Greenland Ice Stream over the 2003–9 period. Two assimilations were attempted, one using surface mass balance as a control, and one using basal friction. In each assimilation, the data consisted of annual elevation change in each year.

Both OpenAD, an open-source AD tool for Fortran (Utke et al., 2008), and the commercial tool TAF (Giering et al., 2005) have been used to differentiate STREAMICE, the ice-sheet component of the MITgcm (Goldberg and Heimbach, 2013; Goldberg et al., 2016), a hybrid Shallow-Shelf, Shallow-Ice model. Using the adjoint model, Goldberg et al. (2015) assimilated a dataset consisting of 10 years of annual surface elevation (with velocities available for 4 of these years) for Smith Glacier, a small but fast-thinning marine glacier in West Antarctica. Surface elevation was not available for ice shelves, and the control set consisted of basal friction parameters and the stress tensor along the 1996 grounding line (the latter representing buttressing from the shelves). In contrast with Larour et al. (2014), the controls were either constant in time, or varied linearly in time over the assimilation period. It was found that time-constant controls were sufficient to reproduce the majority of grounding-line variability over the period, and that allowing the controls to vary linearly in time led to marginal reduction of the cost function. The results suggested that observed 2001–11 retreat was due to the state of the glacier in 2001 rather than ice-shelf melting.

Larour et al. (2014) used controls that varied on the model time step. In other words, the number of degrees of freedom to describe the control was $N_{\Omega} \times N_t$, where N_{Ω} is the number of degrees of freedom of a finite-element function on the model grid and N_t is number of (monthly) time steps – giving a control dimension far larger than that of the observations. While this demonstrates the power of the adjoint in efficient derivative calculation, the findings of Goldberg et al. (2015) suggest that a low number of temporal degrees of freedom should be investigated first to avoid over-fitting.

Assimilation of Internal Layers

Time-dependent assimilation has also been implemented in the context of inferring past climate and ice-sheet behaviour from internal ice layers. While distinct from the applications of Larour et al. (2014) and Goldberg et al. (2015) in terms of investigative questions and types of data utilised, the computational approach is similar.

Internal reflecting horizons (IRHs) in ice sheets are caused by buried impurities deposited on the surface in past epochs, and can often be traced via airborne ice-penetrating radar over large expanses (Karlsson et al., 2014). As material surfaces, they reflect past deformation, as well as surface and basal mass balance (Bell et al., 2011), and, coupled with borehole data, can provide age constraints, and potentially knowledge of past flow, climate, and basal conditions (Bodart et al., 2021). Age-depth information has been used to infer Holocene surface accumulation rates (e.g. Bodart et al., 2021) as well as records of historic changes in flow direction and speed (e.g. Conway et al., 2002; Siegert et al., 2013). These and similar studies, however, used methods that are either qualitative in nature or did not take heterogeneity in ice-sheet flow into account (Dansgaard et al., 1969).

Waddington et al. (2007), by contrast, minimised the model–data misfit to infer accumulation rate patterns, layer ages, and basal conditions. The forward model of Waddington et al. (2007) is a steady-state thermomechanical Shallow-Ice model in a flow-band ($x - z$) domain, which provides surface elevation and surface velocity as observables. In addition, the model implements a Lagrangian particle-tracking method for particle position (x, z) at time t :

$$\begin{bmatrix} x(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} x_0 \\ s(x_0) \end{bmatrix} + \int_0^t \mathbf{v}(x, z, t') dt', \quad (6.40)$$

where $\mathbf{v} = (v_x, v_z)^T$ is the ice velocity, and s is the steady surface elevation profile. Equation (6.40) is used to find profiles of layers of a given age. The authors used this model to minimise the misfit to observed surface elevation and velocity as well as layer geometry, using surface mass balance and layer age as controls – providing a time-constant estimate of accumulation. The method was further developed to allow for a time-evolving model (Koutnik and Waddington, 2012), allowing for assimilation of ice-core and radar layer data in the West Antarctic Ice Sheet divide (Koutnik et al., 2016).

Very often, ages are not available for radar layers, making it challenging to infer past ice-sheet conditions. The model of Hindmarsh et al. (2009) uses three-dimensional velocities from a steady Shallow-Ice approximation model to solve the age-depth equation:

$$\frac{DA}{Dt} = \frac{\partial A}{\partial t} + \mathbf{v} \cdot \nabla A = 1, \quad (6.41)$$

where A is age (and A is set to zero at the surface $z = s$) and DA/Dt is its material derivative. A sophisticated least-square fitting algorithm is then used to identify isolines of modelled A , which best fit observed layers. Karlsson et al. (2014) used this model to evaluate the fit of potential Holocene mass balance scenarios using three-dimensional layer data from Pine Island Glacier.

To date, a formal assimilation of englacial radar data with a continental-scale, three-dimensional dynamic ice flow model has not been carried out. Such assimilation requires cost function gradient information in order to be tractable. In Waddington et al. (2007), the simplicity and limited spatial scale of the model allowed the gradients to be calculated simply by perturbing individual controls to assess cost function partial derivatives, but this finite-difference approach would become intractable for a large-scale study. However, Heimbach and Bugnion (2009) and Logan et al. (2020) successfully applied OpenAD to the ice-sheet model SICOPOLIS (Greve et al., 2011), a thermomechanical ice-sheet model used for studies over paleo-time scales, which implements layer-tracing algorithms. Combined with an increasingly comprehensive set of dated IRH's in the Antarctic and Greenland Ice Sheets (Ashmore et al., 2020;

MacGregor et al., 2015), there is potential for assimilation of ice-sheet histories to now take place.

6.3 Ensemble and Probabilistic Approaches

The methods discussed in Section 6.2 were developed with the aim of minimising a cost function (i.e. of finding an optimal set of parameters to obtain model–data fit). In general, they do not provide formal uncertainties for these parameters – uncertainties which are important to know, especially if the parameters, or the optimised models, are used for analysis or projection. Obtaining uncertainties for inverted parameter sets of large dimension is a challenging problem, and methodologies are only beginning to be developed for glaciological assimilation. In this section, the leading methods are briefly discussed.

6.3.1 Bayes' Theorem

Most probabilistic approaches can be understood as an implementation of Bayes' theorem, which describes how new information changes the likelihood of hidden parameters. Bayes' theorem states that, for a parameter α , and data $\mathbf{y}^{obs}$,

$$p(\alpha|\mathbf{y}^{obs}) = \frac{p(\mathbf{y}^{obs}|\alpha)p(\alpha)}{p(\mathbf{y}^{obs})}. \quad (6.42)$$

With this formula, the *posterior* probability distribution $p(\alpha|\mathbf{y}^{obs})$ – that is, the likelihood of the parameters conditioned on what is observed – can be quantified. Different probabilistic approaches to assimilation make use of Eq. (6.42) in different ways. In the case of VDA, this leads to a continuous function to be optimised for maximum likelihoods (see Section 6.3.3); while in Monte Carlo analysis (see Sections 6.3.4 and 6.4.3), it is used to directly sample the posterior, and in ensemble methods it is used to assign ‘likelihoods’ to members of large ensembles. These methods are explored in the following.

6.3.2 Bayesian Calibration via Ensemble-Based Emulation and Dimensional Reduction

As mentioned, Bayesian calibration can be applied to large ensembles of model output, generated by varying key parameters. In such approaches, an ensemble $\{\alpha_1, \dots, \alpha_N\}$ is generated, where α_i represents a specific realisation of model parameters and its associated model output. Likelihoods are assigned based on a probability distribution of observables. For instance, if a model output or set of outputs yields unlikely results, the likelihoods of the corresponding parameter realisations are scaled by this small probability. It is generally assumed that the parameter realisations are uniformly distributed (i.e. that the prior

distribution $p(\alpha)$ is uniform). Ritz et al. (2015) generated a 1,000-member ensemble of Antarctic-wide simulations from 2000–2200 with GRISLI, a three-dimensional thermo-mechanical Shallow-Ice model (Quiquet et al., 2018) with a set of 13 parameters governing parameterised marine ice retreat. Likelihoods were assessed based on observed mass loss values from a number of Antarctic catchments and their uncertainties, in order to generate a posterior distribution for Antarctic ice loss over the next two centuries.

The ensemble approach can be computationally expensive when using regional or continental scale models, particularly if the models implement higher-order stress balances or use a high spatial resolution to capture phenomena such as marine ice retreat. Moreover, an ensemble generated *a priori* may not have sufficient resolution in parameter space in the neighbourhood of the point of maximum likelihood (sometimes referred to as the *maximum a posteriori*, or MAP point). For this reason, statistical *emulators* are sometimes used (Edwards et al., 2021). An emulator takes $\tilde{\mathbf{Y}}$, an m -by- n matrix in which the i _{th} column contains model outputs (of dimension m) from ensemble member i (i.e. for the given set of parameters α_i), and carries out a singular value decomposition:

$$\tilde{\mathbf{Y}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T, \quad (6.43)$$

where $\mathbf{U}$ and $\mathbf{V}$ are unitary matrices, and $\mathbf{\Sigma}$ is a diagonal matrix containing the *principal values* of the ensemble. All but the largest values of $\mathbf{\Sigma}$ are set to zero, resulting in a mapping between parameter values α_i and a low-dimensional representation of model output. (In Wernecke et al., 2020), five principal values are retained.) This discrete mapping is then statistically interpolated (e.g. with a Gaussian process model) to provide a continuous mapping, which can be queried in a Bayesian calibration based on observations. Rather than using aggregated data as in Ritz et al. (2015), Wernecke et al. (2020) project observational data into principal spatial components for comparison with emulator output.

6.3.3 Bayesian Inference with Laplacian Approximation

Ensemble methods are somewhat limited by the small number of parameters, ($\mathcal{O}(\sim 10)$), that can be considered. As discussed in Section 6.2, the parameters that determine ice-sheet dynamics are spatial fields, requiring, in general, a very large (i.e. $10^4 - 10^6$) number of degrees of freedom to describe them. While these parameter fields likely exhibit spatial correlations, effectively reducing their dimensionality, their correlation scales are unknown, preventing *a priori* dimension reduction. However, there is a connection between the variational data assimilation described in Section 6.2 and Bayesian methods, which can be exploited. Assume a model, which calculates ice-sheet surface velocities, $\mathbf{v}$, based on parameters $\mathbf{C}$, and there are observations, $\mathbf{v}^{obs}$,

for which error is normally distributed. In the following, observations are treated as measurements at discrete locations and described by a vector (although of different dimension than the control parameters $\mathbf{C}$).

The likelihood operator (cf. Eq. (6.42)) can be expressed as

$$p(\mathbf{v}^{obs}|\mathbf{C}) \propto e^{-\frac{1}{2}|\mathbf{v}^{obs}-\mathcal{H}(\mathbf{v}(\mathbf{C}))|_{\Gamma_{obs}^{-1}}^2}, \quad (6.44)$$

where $\mathcal{H}$ is the observation operator (introduced in Section 6.1.3). Here the constant of proportionality is simply a normalisation constant which does not bear on the analysis, and which we subsequently ignore. We can equivalently consider the negative log-likelihood:

$$-\log(p(\mathbf{v}^{obs}|\mathbf{C})) = \frac{1}{2}|\mathbf{v}^{obs}-\mathcal{H}(\mathbf{v}(\mathbf{C}))|_{\Gamma_{obs}^{-1}}^2. \quad (6.45)$$

Here, Γ_{obs} is the *observational covariance matrix*, and the norm $|\mathbf{x}|_{\Gamma_{obs}^{-1}}$ is defined as $\sqrt{\mathbf{x}^T \Gamma_{obs}^{-1} \mathbf{x}}$. If the prior distribution $p(\mathbf{C})$ is also characterised as a multivariate Gaussian with covariance Γ_{prior} and mean $\mathbf{C}_0$, the negative log of the posterior distribution can be written

$$-\log(p(\mathbf{C}|\mathbf{v}^{obs})) = \frac{1}{2}|\mathbf{v}^{obs}-\mathcal{H}(\mathbf{v}(\mathbf{C}))|_{\Gamma_{obs}^{-1}}^2 + \frac{1}{2}|\mathbf{C}-\mathbf{C}_0|_{\Gamma_{prior}^{-1}}^2. \quad (6.46)$$

Note that both Γ_{prior} and Γ_{obs} are symmetric positive-definite matrices.

It can be shown (e.g. Isaac et al., 2015) that finding the point in parameter space with maximum posterior likelihood, $\mathbf{C}_{MAP}$, is equivalent to variational data assimilation with the appropriate data and model norms. If Γ_{obs} is diagonal, then the first term on the right-hand side of Eq. (6.46) is equivalent to the misfit norm (Eq. (6.8)).

Additionally, the second term on the right-hand side of Eq. (6.46) corresponds to the discretised form of a wide range of commonly used regularisation cost terms, or model norms. For instance, if a finite-element model is used and $\mathbf{C}_0$ is spatially invariant, the regularisation cost term in Eq. (6.10) can be written

$$\gamma \int_{\Omega} |\nabla C|^2 dx = (\mathbf{C}-\mathbf{C}_0)^T \mathbf{A} (\mathbf{C}-\mathbf{C}_0), \quad (6.47)$$

where $\mathbf{C}$ is a vector representing the coefficient vector of the finite-element function C and a_{ij} , the entries of $\mathbf{A}$, are $\int_{\Omega} \gamma \nabla \varphi_i \cdot \nabla \varphi_j d\Omega$, with $\varphi_{i,j}$ the finite-element basis functions. Note that $\mathbf{A}$ as defined here is not (the inverse of) a true covariance matrix, as it is not positive definite. Barnes et al. (2021) define the regularisation term for the U' a ice model as

$$\int_{\Omega} \gamma |\nabla (C-C_0)|^2 + \delta (C-C_0)^2 dx. \quad (6.48)$$

When discretised, this term expresses a proper prior distribution of $\mathbf{C}$. In such a prior, δ expresses our

confidence in the value of $\mathbf{C}$ at a location, while γ expresses our confidence in its persistence over a certain length scale. This regularisation term uses a gradient operator, but such an operator leads to infinite pointwise variance (Bui-Thanh et al., 2013), and so methods which quantify posterior uncertainty with Eq. (6.46) use Laplacian-based regularisation (Isaac et al., 2015).

The equivalence between cost function and negative-log posterior density (with the caveat that the regularisation term must be a proper probability distribution) suggests that minimising a deterministic function with respect to the control parameters is equivalent to finding the point of maximum likelihood, or *mode*, of the posterior distribution. This alone does not aid in quantifying the uncertainty of $\mathbf{C}$, however: the probability space is of very high dimension (tens to hundreds of thousands in realistic applications) and querying the posterior density requires solving a large-scale non-linear model (Martin et al., 2012). A *local* approximation of the posterior density can be found through a second-order approximation of Eq. (6.46) in the neighbourhood of $\mathbf{C}_{MAP}$. It can be shown that the covariance of this local approximation is given by the inverse of the *Hessian* of the cost function at $\mathbf{C}_{MAP}$:

$$\Gamma_{post}^{-1} = \mathbf{H}(\mathbf{C}_{MAP}) = \frac{\partial^2 \mathcal{J}}{\partial \mathbf{C}^2} |_{\mathbf{C}_{MAP}} = \mathbf{H}_{mis}(\mathbf{C}_{MAP}) + \Gamma_{prior}^{-1}, \quad (6.49)$$

where $\mathbf{H}_{mis}(\mathbf{C}_{MAP})$ is the Hessian of the data norm only. Essentially, the approach approximates $p(\mathbf{C}|\mathbf{v}^{obs})$ by $\mathcal{N}(\mathbf{C}_{MAP}, \mathbf{H}(\mathbf{C}_{MAP})^{-1})$, that is, a Gaussian distribution with mean $\mathbf{C}_{MAP}$ and covariance $\mathbf{H}(\mathbf{C}_{MAP})^{-1}$. This approximation is sometimes referred to as a *Laplacian* approximation.

Calculating the complete Hessian is challenging as it is a large, dense matrix requiring extensive computation, and inverting directly is even more challenging as it is often singular or near-singular. Previous authors (Bui-Thanh et al., 2013; Petra et al., 2014; Isaac et al., 2015) find a low-rank approximation to the Hessian that is easy to invert (see also Kalmikov and Heimbach, 2014). Details can be found in these and other references, but the low-rank approximation is generated via a singular value decomposition of the *prior-preconditioned* misfit Hessian, $\Gamma_{prior} \mathbf{H}_{mis}$. The space of $\mathbf{C}$ is decomposed into principal components – with the significance that the leading principal components, or directions, are those most strongly constrained by the assimilation (Kozioł et al., 2021).

Together with an adjoint of the forward model, the parameter uncertainty Γ_{post} can be projected to estimate uncertainties related to various scalar Quantities of Interest (*QoIs*), such as future sea level rise contribution. Using properties of joint Gaussian distributions,

$$\sigma_{QoI}^2 = \left(\frac{\partial QoI}{\partial \mathbf{C}} \right)^T \Gamma_{post} \left(\frac{\partial QoI}{\partial \mathbf{C}} \right), \quad (6.50)$$

where σ_{QoI}^2 is the variance of the Quantity of Interest. (σ_{QoI} , the standard deviation, is then a proxy for uncertainty). Isaac et al. (2015) used this framework to estimate posterior uncertainty of sliding parameters in a full-Stokes model of Antarctica, and the resulting uncertainty in ice loss. In this work, their parameter dimension is approximately 1.2 million, but only ~1,000 principal components are retained in their low-rank approximation. The framework represents a powerful method to deal with uncertainty with a parameter space of very high dimension, but it is computationally challenging for a number of reasons, which we discuss in Section 6.3.6. The discussion here emphasises the connection between cost functions and posterior densities. However, it is important that the observational covariance be taken into account correctly. Equation (6.45) is an inner product involving modelled and observed values at discrete locations, and these locations do not depend on model resolution. On the other hand, Eq. (6.30) is a functional which considers observations to be a continuous function, and many inversions use integral-based cost functions. Such an approach introduces mesh-dependent terms into the posterior covariance (Kozioł et al., 2021). Essentially, treating such a cost function as a likelihood function implies a very high density of independent observations, effectively adding information that does not exist and therefore underestimating parameter uncertainty.

6.3.4 Stochastic Newton MCMC Methods

The posterior distribution of parameter uncertainty, Eq. (6.46), is non-Gaussian (even if the observational and prior distributions are Gaussian). As such, the Hessian-based uncertainty quantification (UQ) framework described in Section 6.3.3 is an approximation, and potentially a poor one in the case of highly skewed distributions. Markov-Chain Monte Carlo (MCMC) methods provide a means for sampling the posterior density, but the high-dimensional probability spaces and computational expense of the forward model make this challenging. Stochastic Newton MCMC methods build on the Hessian-based methods discussed in Section 6.3.3 to make this more tractable.

Markov-Chain Monte Carlo methods generate new samples of a probability space by selecting based on a *proposal density* centred at the previous sample. For a given sample $\mathbf{a}_t$, Stochastic Newton MCMC uses as its proposal density the local approximation to the posterior given by

$$\mathcal{N}(\mathbf{a}_t, \mathbf{H}(\mathbf{a}_t)^{-1}). \quad (6.51)$$

Importantly, the evaluation of the posterior density is not approximated – only the basis on which new samples are

taken. The low-rank approximation discussed in Section 6.3.3, which retains the directions in probability space which are most constrained by the assimilation, ensures these directions are explored.

Stochastic Newton MCMC has been shown to accelerate ‘standard’ MCMC schemes by two orders of magnitudes in seismic inversion problems (Martin et al., 2012). However, it requires a new, Hessian-based approximation to the posterior for each new sample, which is computationally costly. The approach can be modified, however, by computing the Hessian only at the MAP point, and using the corresponding density as a proposal distribution, rather than recomputing the proposal density with every sample. It was found that using the MAP-based Hessian led to convergence at least as rapidly as the original Stochastic Newton MCMC method.

The problems considered in Martin et al. (2012) and Petra et al. (2014) were of large dimension ($\mathcal{O}(\sim 100)$) relative to the ensemble-based methods discussed in Section 6.3, but quite small compared to adjoint-based inversions in glaciology (e.g. Cornford et al., 2015). Thus, more work is needed to determine whether such methods are tractable for continental ice-sheet models.

6.3.5 Ensemble Kalman Filter Approach

The time-dependent variational approach in Section 6.2.3 has similarities to a variational assimilation approach developed in numerical weather prediction (NWP), 4DVAR. Another method used in NWP is the EnKF (Lorenc, 2003), essentially a Monte Carlo analysis in which an ensemble of model states are run forward in time. The states are updated at discrete time intervals based on observational data.

Gillet-Chaulet (2020) applied EnKF techniques to glaciological assimilation. Only the basic idea of the algorithm is given here; for further details, please see Gillet-Chaulet (2020). The forward model used in this study is a Shallow-Shelf Approximation. The model ‘state’ considered is composed of (time-dependent) ice surface elevation and (time-independent) bed elevation and basal friction, and observations are of surface elevation and surface velocity, available at times t_k . An ensemble of state vectors $\{\mathbf{x}\}$ is generated, where $\mathbf{x}_i$ represents the discretised fields comprising the state for realisation i . The filter then consists of a sequence of forecast-analysis steps. Each member of $\{\mathbf{x}\}$ is evolved from time t_{k-1} to time t_k , yielding the *forecast* $\{\mathbf{x}\}^{f,k}$, the *forecast* ensemble at time t_k . An update is then applied to generate the *analysis* ensemble:

$$\mathbf{S}^{a,k} = (\mathbf{B}^T \Gamma_{obs} \mathbf{B} + \mathbf{S}^{f,k})^{-1}, \quad (6.52)$$

$$\bar{\mathbf{x}}^{a,k} = \bar{\mathbf{x}}^{f,k} + \mathbf{S}^{a,k} \mathbf{B}^T (\Gamma_{obs}^k)^{-1} (\mathbf{y}_{obs}^k - \mathcal{H}(\mathbf{x}^{f,k})). \quad (6.53)$$

Here $\bar{\mathbf{x}}^{f,k}$ and $\mathbf{S}^{f,k}$ are the mean and covariance matrix of the forecast ensemble, $\bar{\mathbf{x}}^{a,k}$ and $\mathbf{S}^{a,k}$ similarly for the analysis, and (Γ_{obs}^k) is the observational covariance at t_k . The

subglacial hydrology and higher-order glacier physics – and therefore directly model basal slipperiness rather than representing it with a high-dimensional parameter space.

The methods all consider low-rank approximations in order to make their computations tractable. Ensemble and EnKF methods carry out principal component decompositions on computed ensembles, meaning the methods ultimately depend on the information contained within the ensembles, which must be generated prior to the analysis. By contrast, Hessian-based UQ and Stochastic Newton MCMC prioritise the directions of parameter space that are most strongly determined by the data. Thus, the components analysed are, by design, the most important to consider.

Ensemble and EnKF methods do not require extensive modifications to the ice-sheet models themselves. Emulation-based ensemble studies require only the output of a model ensemble, and in EnKF methods the analysis computations can be run in an ‘offline’ mode, where the forecast state is processed externally to provide the analysis state. Hessian-based UQ and Stochastic Newton MCMC for non-linear models require second-derivative information; such information requires either higher-order AD (Kalmikov and Heimbach, 2014; Maddison et al., 2019) or the solution of differential equations not solved by most ice-sheet models (Petra et al., 2012; Isaac et al., 2015). Applying AD remains a challenge for established ice-sheet models, with successes primarily only where the model code adheres to a certain structure (Goldberg and Heimbach, 2013; Goldberg et al., 2016; Logan et al., 2020). There is growing use of automated code generation finite-element libraries such as FEniCS and Firedrake in glaciological model development (Kyrke-Smith et al., 2017; Brinkerhoff et al., 2021), and features of these libraries can easily be exploited to find higher-order functional derivatives (Maddison et al., 2019; Koziol et al., 2021). It also may be sufficient in certain cases to avoid high-order derivative calculation by using the Gauss–Newton approximation to the Hessian (Koziol et al., 2021; Shapero et al., 2021):

$$\mathbf{H}_{mis} \approx \mathbf{B}^T \Gamma_{obs}^{-1} \mathbf{B}, \quad (6.54)$$

where $\mathbf{B}$ is the gradient of the observation operator (cf. Section 6.3.3). Essentially, AD and derivative calculation for probabilistic ice-sheet data assimilation is not as great a barrier as it once was.

All of the methods discussed here have strengths and weaknesses and the ideal method will depend on the problem being investigated. The way forward may lie not in choosing a method, but in combining different methods. For instance, elements of Hessian-based UQ and Stochastic Newton MCMC may aid in generating ensembles for EnKF which efficiently capture variability near the mode of the posterior and enable smaller ensembles.

6.3.7 Alternative Approaches

Many of the issues mentioned in Section 6.3.6 (non-Gaussianity, computational complexity) arise because of the non-linearity of the glaciological flow equations. A completely different approach has been proposed by Raymond and Gudmundsson (2009), which makes use of a linear theory developed by Gudmundsson (2003) to estimate basal properties of glaciers using a probabilistic Bayesian approach. In this approach, basal topography and slipperiness are inferred from surface elevation and velocity. However, due to the linear approximations made, the method is limited to small perturbations around a known reference state. The framework is not suitable for constraining projection uncertainty but may be effective in characterising basal environments. To date, the framework has been applied to a flowline on Rutford Ice Stream (Pralong and Gudmundsson, 2011).

6.3.8 Model Uncertainty

The likelihood operator as written (Eq. (6.45)), implicitly assumes that the model $\mathbf{v}(\boldsymbol{\alpha})$ is *perfect*, and that any uncertainty is due to imperfect observations. In truth, there is a great degree of *model uncertainty* (i.e. error in mathematically and numerically representing the physical system). Model uncertainty is extremely difficult to characterise (Chatfield, 1995), and various studies take different approaches. The idealised studies of Gillet-Chaulet (2020) and Koziol et al. (2021) ignore model uncertainty completely; Wernecke et al. (2020) consider model uncertainty as uncorrelated Gaussian noise based on the variance inherent in their ensemble; and Werder et al. (2020) (see Section 6.4.3) treats per-glacier model uncertainty as additional parameters in the Bayesian estimation.

Babaniyi et al. (2021) applied a Bayesian Approximation Error (BAE) approach to estimate ice-sheet basal properties in the presence of rheological uncertainty. Ice-sheet rheology is governed by parameters which are themselves uncertain (such as the Glen’s law exponent, n). The BAE step involves generating an ensemble of model output by sampling from prior distribution parameters, and using the ensemble statistics to produce a modified likelihood covariance matrix which accounts for the impacts of rheological uncertainty. The BAE framework could, in theory, be applied to a range of structural uncertainties in ice-sheet models, leading to a more accurate accounting of model uncertainty – but this, as well as application to realistic problems, still needs to be done.

6.4 Reconstructing of Bed Topography and Ice Thickness

Data assimilation methods have also been employed to map the beds of glaciers and ice sheets. Ice thickness and bed topography are most efficiently measured by ice-penetrating

radar (either ground-based or airborne; Evans and Robin, 1966), but the vast majority of mountain glaciers, and many parts of the ice sheets remain unmapped. Data assimilation offers a cost-effective way to construct estimates where no data are available. Several approaches have been proposed (see, e.g., the Ice Thickness Models Intercomparison eXperiment projects; Farinotti et al., 2017, 2021), and we will only focus on a few popular ones. Most approaches are based on the conservation of mass (Eq. (6.6)), the conservation of momentum (Eq. (6.1)) or a combination of the two.

6.4.1 Two-Step Methods

The majority of bed/thickness mapping methods rely on a two-step approach, which consists of using the conservation of mass first, followed by the conservation of momentum.

Flux Conservation

In the first step, we rely on the mass transport equation to determine the depth-integrated flux $\mathbf{q} = q \mathbf{n} = H\bar{\mathbf{v}}$:

$$\nabla \cdot q \mathbf{n} = \dot{m}_s - \dot{m}_b - \frac{\partial H}{\partial t} = \dot{a}, \quad (6.55)$$

where $\mathbf{n}$ is a unit vector describing the two-dimensional direction of the depth-integrated flux, q , and $\dot{a}$ is the right-hand side, sometimes referred to as *apparent mass balance*. If the right-hand side and the flux directions are known, the distribution of the flux can be determined by using a routing algorithm (e.g. Budd and Warner, 1996) or by solving this transport equation using standard approaches such as finite-elements or finite-differences.

To estimate the flux directions, one can use surface gradients, as dictated by the Shallow-Ice Approximation. This approximation will only be valid at a length scale of a few ice thicknesses and some smoothing of the surface topography must be applied. Another possible approach is to use observed surface velocities instead, but many mountain glaciers remain poorly mapped.

For the right-hand side, the basal melt term is generally negligible under grounded ice compared to the other terms. The surface mass balance can come from regional climate models but is, most of the time, simply parameterised as a function of the elevation with respect to the Equilibrium-Line Altitude (ELA), the presence of debris and other parameters (e.g. Farinotti et al., 2009). The thinning rate is generally ignored, but can be estimated from surface altimetry.

Momentum Balance

The flux conservation step yields a complete map of the depth-integrated flux $q = H|\bar{\mathbf{v}}|$. In order to reconstruct the ice thickness and bed topography, an estimate of the depth-averaged velocity $\bar{\mathbf{v}} = |\bar{\mathbf{v}}|$ is needed. The vast majority of bed

mapping methods for mountain glaciers relies on the Shallow-Ice Approximation (SIA; Hutter, 1983), which is a zeroth-order approximation of ice sheet flow and only accounts for vertical shear stresses. Under the SIA assumptions, the ice velocity reads:

$$\mathbf{v}(z) = \mathbf{v}_b - 2(\rho g)^n |\nabla s|^{n-1} \nabla s \int_b^z (s - z)^n A \, dz, \quad (6.56)$$

where $\mathbf{v}_b$ is the basal sliding speed, and s is the surface elevation. If it is assumed that ice is isothermal (i.e. A is uniform with depth), and that the ice basal velocity is negligible ($\mathbf{v}_b = \mathbf{0}$), the ice thickness can be reconstructed as:

$$H = \left(\frac{(n+2)q}{2A(c\rho g)^n |\nabla s|^n} \right)^{1/(n+2)}, \quad (6.57)$$

where c is a parameter introduced to account for potential sliding. This method, or similar approaches, have been used in many studies (e.g. Fastook et al., 1995; Warner and Budd, 2000; Farinotti et al., 2009; Huss and Farinotti, 2012) and led to the first global mapping of mountain glacier volumes (Huss and Farinotti, 2012).

Fürst et al. (2017) proposed a more advanced approach, where the surface mass balance is first used as a control in Eq. (6.55) in order to have a flux that is positive and smooth, and use the SIA to infer the bed topography. The flow rate factor is calibrated with radar-derived thickness measurements. Then, in a second step, a second optimisation is performed in regions where surface velocities are available. In these regions, another cost function is minimised that measures the misfit between the calculated ice thickness and radar measurements (together with other terms to ensure that the inferred ice thickness is physically consistent). This second optimisation is only performed in regions of fast flow, where ice flow is primarily controlled by basal sliding, and the controls are the surface mass balance and ice velocities. This approach was first applied to Svalbard (Fürst et al., 2018).

6.4.2 Mass Conservation with Observed Velocities

There is now complete coverage of surface velocities for both ice sheets and several large ice caps from satellite interferometry (Rignot et al., 2011; Rignot and Mouginot, 2012; Joughin et al., 2018). In this case, one can rely directly on these observations instead of the SIA. However, assumptions are needed regarding how to convert these surface velocities to depth-averaged velocities. In some studies (Rasmussen, 1988; Morlighem et al., 2010), it is assumed that surface velocities are good approximations for depth-averaged velocities. In the worst-case scenario, when there is no basal sliding, Eq. (6.56) shows that depth-averaged velocities can be 20% smaller than surface velocities ($\bar{\mathbf{v}} = (n+1)/(n+2) \mathbf{v}(s)$). In regions of fast flow, far from the ice divide and along ice streams, this is an excellent

approximation that avoids having to introduce basal conditions and rheological factors. In this case, the ice thickness is estimated by solving for ice flux (Eq. (6.55)), where the ice thickness is the unknown (i.e. the velocity and velocity directions are given). This hyperbolic equation requires measurements of ice thickness at the inflow boundary, Γ_- , of the domain in order to properly constrain the system:

$$\begin{cases} \nabla \cdot H \bar{\mathbf{v}} = \dot{a} & \text{in } \Omega \\ H = H_{obs} & \text{on } \Gamma_- \end{cases} \quad (6.58)$$

Morlighem et al. (2011, 2013a) further refined this method by assimilating radar-derived ice thickness measurements in order to constrain the calculation so that the final bed map is as close as possible to observations where they are available. To account for all measurements of ice thickness H_{obs} , along flight tracks T , that lie within the model domain Ω , the following cost function is minimised:

$$\begin{aligned} \mathcal{J}(H) = & \int_T \frac{1}{2} (H - H_{obs})^2 dT + \int_{\Omega} \gamma_{\parallel} (\nabla H \cdot \mathbf{n}_{\parallel})^2 d\Omega \\ & + \int_{\Omega} \gamma_{\perp} (\nabla H \cdot \mathbf{n}_{\perp})^2 d\Omega, \end{aligned} \quad (6.59)$$

where $\mathbf{n}_{\parallel}$ and $\mathbf{n}_{\perp}$ are unit vectors parallel/perpendicular to ice velocity, and $\gamma_{\parallel}$ and $\gamma_{\perp}$ are constant regularisation parameters. The controls are the observed velocity and the apparent mass balance. These input fields are allowed to change within their respective error margins. This method led to the development of BedMachine Greenland (Morlighem et al., 2014, 2017) and BedMachine Antarctica (Morlighem et al., 2020).

6.4.3 Bayesian Method

Werder et al. (2020) proposed to combine the two-step approach from Huss and Farinotti (2012) with a Bayesian inversion scheme. In Werder et al. (2020), the parameters α (cf. Eq. (6.42)) are apparent mass balance (parameterised in Huss and Farinotti, 2012), ice temperature, basal sliding factor, and extrapolation parameters for thickness and velocity. The last two parameters are considered because the method of Huss and Farinotti (2012) collapses each glacier into a flowline, and then extends to a thickness map through an ad-hoc extrapolation. The observations considered are surface velocities (which the model of Huss and Farinotti, 2012 was augmented to calculate) and (where available) thickness measurements. A Markov-Chain Monte Carlo (MCMC) method is used, allowing for efficient investigation of the posterior probability density. The method is very powerful as it does not presume the form of the posterior distribution (as opposed to methods which make the Laplacian approximation) and allows a means to incorporate the growing repository of glacier velocities. However, the parameters are not spatially resolved within a glacier due to computational expense, even with the use of MCMC.

6.4.4 Simultaneous Mass and Momentum Method

Perego et al. (2014) introduced a variational assimilation method, which infers both the basal sliding parameter C (c.f. Section 6.2) and ice-sheet topography. The method uses an approach similar to that described in Section 6.2.2 – with key differences being that the cost function includes a term penalising any imbalance in the mass-transport equation, and also that ice thickness (or equivalently ice bed) is included as a control parameter. The method was developed primarily to initialise climate-forced ice-sheet models without large non-physical transients in thickness evolution, and was successfully applied to a model of the Greenland Ice Sheet.

As with other ‘dual’ inversions for multiple parameters (Section 6.2.2), there is risk of parameter mixing and non-uniqueness of solution (Goldberg and Heimbach, 2013). For this reason, it may be a preferable strategy to adopt an approach for bed inversion which does not depend on the momentum balance (such as mass continuity) and then use the resulting topography to infer ice-dynamical parameters.

6.4.5 Artificial Intelligence and Machine Learning

Neural networks are being used more and more frequently in environmental science to improve the computational efficiency of certain processes in numerical model (Krasnopolsky and Schiller, 2003), or to infer unknown quantities from a set of input data.

One of the first applications of neural networks in glaciology is from Clarke et al. (2009), who used a neural network to infer the subglacial topography of mountain glaciers from surface topography with some success.

More recently, Leong and Horgan (2020) proposed a new technique that is based on adapted architecture of the enhanced super-resolution Generative Adversarial Network (GAN) to reconstruct basal roughness in bed topography maps from ice surface elevation, velocity, and snow accumulation. The neural network is trained in a few regions of Antarctica where high-resolution (250 m) bed elevation maps are available, and applied over the entire ice sheet to generate high resolution bed topography. Meanwhile Jouvett et al. (2022) and Jouvett (2023) introduce an approach, which a deep learning-based ice dynamics emulator acts as the forward model in the assimilation of observations. The emulator is trained on the output of an ensemble of runs carried out with a physically-based glacier model. These sorts of new approaches have a lot of potential as the amount of remote sensing data collected in polar regions increases.

6.5 Perspectives and Outlook

While data assimilation has been an important and popular research topic since the 1990s, glaciology has been rapidly shifting from a data-poor state to a data-rich one. In the 1990s, modellers only had partial coverage of ice velocity

and the thinning rates were mostly unknown. Data assimilation was a powerful tool to add knowledge to the model, by inferring boundary or initial conditions. Today, we have time series of complete velocity maps and thinning rate maps at an ever-increasing spatial and temporal resolution. Modellers are not yet equipped to take advantage of this data revolution, as the vast majority of glaciological assimilation platforms cannot assimilate this breadth of information. There may additionally be data redundancies (e.g. ice velocities may be of such high resolution that more resolved velocities do not add information); however, without sophisticated uncertainty quantification, these will go undetected. In order that both existing and newly collected data are used to their full potential in improving projections of sea level, transformative data science and machine learning solutions may be needed.

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